From one bit to a fitted polynomial: engineered spectral filters for track reconstruction in a noisy LHCb VELO toy model

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July 18, 2026

Abstract

Track reconstruction in the LHCb vertex detector can be written as a linear system $A\mathbf{x} = \mathbf{b}$ over track *segments*, whose solution scores each segment by how strongly its neighbours reinforce it. Two earlier papers solved this system with quantum linear algebra: first with the HHL algorithm, and then far more cheaply with a one-bit variant, the one-bit quantum filter (1BQF), which replaces the full inversion by a single cosine filter of the Hamiltonian with one spectral zero. In this paper I develop that idea further. The 1BQF is the degree-one member of the family of polynomial spectral transforms provided by the quantum singular value transformation (QSVT), and once the problem is viewed this way the design question changes: it is no longer a question of which algorithm inverts A, but of which function of λ separates true from false segments. On clean events the answer is a four-line comb pinned to the eigenvalues of a five-hit track, and it drives the segment false rate from 20.5% (classical, T=1000 tracks) to 0.9% at 92% efficiency. On noisy events the answer must be measured: the false weight piles up on the eigenvalue lines of specific two- and three-segment motifs, and a polynomial fitted with roots at the measured pile-up beats the classical baseline at every attainable efficiency below a sharply defined *fragment floor*, the small population of true fragments that are spectral twins of the false motifs and that no function of λ can separate. I also show that the two natural Hamiltonian modifications inherited from the classical literature behave very differently from what their classical performance suggests. The Denby–Peterson occupancy term, although it helps the classical solver, is actively harmful to any solver that relies on a spectral zero and has no place in the Hamiltonian. The bifurcation (fork) term reads as a no-go when paired with an unfitted filter, and becomes a net win exactly when the polynomial is refitted to the spectrum it produces. The result is a two-part design rule: the operator shapes the spectrum, and the fitted response separates it; what the spectrum provably cannot separate is left, explicitly, to classical track-level information outside the scope of this paper.

1 Introduction

Charged-particle track reconstruction is, at heart, a combinatorial matching problem: a detector hands us a cloud of hits, and we must decide which hits belong together. At the High-Luminosity LHC this cloud becomes dense enough that the matching cost dominates real-time reconstruction budgets, and the tracking community has responded by testing essentially every alternative computing platform: GPUs, FPGAs, neural networks, and, in the line of work this paper continues, quantum computers.

The lineage of this work is short and worth stating precisely. Nicotra et al. [1] showed that track reconstruction in a simplified model of the LHCb vertex detector (VELO) can be phrased

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as a linear system $A\mathbf{x} = \mathbf{b}$ over track segments and solved with the Harrow–Hassidim–Lloyd algorithm [2]. The follow-up paper [3] observed that the full inversion is overkill. The interesting information is which segments are active, not their precise weights, so a single-ancilla circuit, the one-bit quantum filter (1BQF), suffices. The 1BQF applies one cosine function of the Hamiltonian, placed so that its zero lands exactly on the eigenvalue shared by every isolated fake segment, and reads out the survivors. It is cheap, it suits near-term hardware, and it was demonstrated on trapped-ion and superconducting devices.

This paper asks the question that follows naturally from the 1BQF. A cosine with one zero is a polynomial filter of degree one[1]. The quantum singular value transformation (QSVT) [6, 9] tells us that a quantum computer can apply essentially any bounded polynomial of the Hamiltonian at a cost linear in its degree. So, what could we do with more than one zero? Which polynomial should we choose? And what happens to that choice when the Hamiltonian itself changes, because the detector is noisy or because we have added constraint terms to help it? In my opinion this last question carries most of the physics, and most of this paper is spent answering it.

I answer these questions on the same toy model as the two predecessor papers, extended with realistic noise. The answers can be summarised as a single working method, which I will refer to as fitting the response to the spectrum:

1. On clean events the true-track spectrum is *discrete*: detector geometry pins every true track to the same five-hit chain, whose four eigenvalues are known in closed form. A comb of narrow passes at those four lines, and nothing else, separates true from false essentially perfectly up to the densities I can simulate ($T = 1000$ tracks, four million segments), where the classical solver's false rate has climbed to 20.5%.

2. On noisy events the false population moves: it concentrates on the eigenvalue lines of specific small *motifs* (hit-sharing pairs and triples), which sit off the true lines but inside any naive pass window. The correct response cannot be written down a priori; it must be fitted to the measured pile-up. Once it is, the quantum false rate drops below the classical one at every efficiency below a floor that I can identify, count, and explain segment by segment.

3. Hamiltonian engineering and response engineering are coupled. The occupancy penalty, helpful classically, damages any solver that relies on a spectral zero; I show the mechanism and conclude that the constraint belongs outside the Hamiltonian altogether. The fork penalty genuinely improves the spectrum, but only a refitted response can collect the improvement: with the production comb it reads as a no-go, with the fitted response it is a net win. Changing the operator moves the activations, and moving the activations changes the polynomial we have to fit.

The paper is organised as follows. Section 2 builds the toy model and the segment Hamiltonian from zero, including the exact activation levels that make everything else quantitative. Section 3 reviews the 1BQF as a spectral filter and establishes the working-point conventions used throughout. Section 4 develops the QSVT machinery (block encoding, qubitization, and the linear-combination-of-Chebyshev construction) at the level of detail needed to count qubits and gates. Section 5 designs and evaluates the line comb on clean events, including its resource scaling and its provable limits. Section 6 introduces the noise model and shows what it does to each solver. Section 7 dissects the occupancy term. Section 8 does the same for the fork term and documents the no-go that motivates the refit. Section 9 fits the polynomial to the measured false pile-up and presents the main comparison. Section 10 collects the limitations: realizability, generalization, Trotterization, and what a real detector geometry changes. Section 11 concludes.

A remark on style before we begin. This paper is deliberately verbose. The predecessor papers

[1] Strictly, a trigonometric function realised at degree one in the Chebyshev basis of the walk operator; section 4 makes this exact.

compressed the segment-Hamiltonian formalism to a few paragraphs; here I want every level, every eigenvalue, and every convention on the table, because the physics of why filters succeed and fail lives in exactly those details.

2 The toy model and the segment Hamiltonian

2.1 Events

The toy detector is the standard simplified VELO of refs. [1, 3]: parallel planes perpendicular to the beam axis (z), each recording a two-dimensional point where a charged particle crosses it. An event is generated by drawing T tracks from a common primary vertex on the z-axis, with polar angles inside a generation cone, and propagating each as a straight line through the planes. Every track leaves one hit per plane. Detector effects enter as controlled perturbations, and I will always state the triple in force: multiple-scattering kinks of width σ_{scatt} applied at each plane crossing, Gaussian hit-position smearing of width σ_{res}, and a hit-drop (inefficiency) probability applied hit by hit. The clean configuration used in section 5 is $(\sigma_{\text{scatt}}, \sigma_{\text{res}}, \text{drop}) = (10^{-4}, 0, 0)$; the heavy configuration used from section 6 onward is $(10^{-4}, 20\,\mu\text{m}, 1\%)$.

A *segment* is an ordered pair of hits on adjacent planes. With five planes and T hits per plane, each of the four inter-plane gaps offers T^2 candidate segments, so

$$n_{\text{seg}} = 4T^2, \qquad n_{\text{true}} = 4T, \qquad n_{\text{false}} = 4T(T-1), \tag{1}$$

where n_{true} counts the segments that connect two hits of the same generated track. The problem is a needle in a haystack, and the ratio $n_{\text{true}}/n_{\text{seg}} = 1/T$ says how deep the hay is: at $T = 1000$ only one segment in a thousand is real, and the false population grows quadratically while the true one grows linearly. Any segment classifier, classical or quantum, fights this composition before it fights anything else.

2.2 The Hamiltonian

Following Denby and Peterson's neural formulation of track finding [4, 5] as adapted by refs. [1, 3], we encode an event in a quadratic energy over segment activations $x_i \in \mathbb{R}$,

$$E(\mathbf{x}) = \tfrac{1}{2}\,\mathbf{x}^{\top} A\,\mathbf{x} - \delta\,\mathbf{1}^{\top}\mathbf{x}, \qquad A = (\gamma + \delta)\,I - C, \tag{2}$$

where C is the 0/1 *compatibility adjacency*: $C_{ij} = 1$ exactly when segments i and j share a hit as head-to-tail continuations (the end hit of one is the start hit of the other) and their mutual angle is below an acceptance ε. The constants are a self-penalty γ (discouraging indiscriminate activation) and a bias δ (rewarding activation); throughout the main text $\gamma = 3$, $\delta = 1$, so the constant diagonal is $s \equiv \gamma + \delta = 4$. Minimising eq. (2) gives the linear system

$$A\,\mathbf{x} = \delta\,\mathbf{1}, \tag{3}$$

which I solve classically with MINRES (A is symmetric, sparse, and at $\gamma = 3$ comfortably positive definite). A useful picture is a voting market: each segment starts with the same bias-funded credit δ, and the off-diagonal couplings let aligned neighbours reinforce one another. A segment embedded in a long, straight chain of compatible neighbours accumulates support; a segment with no compatible continuation is left with only its own bias against its own penalty.

The acceptance ε deserves its own sentence, because it is the single most consequential experimental knob. It is set by a closed-form formula from the expected root-mean-square kink between consecutive true segments under the noise triple in force, so that true continuations pass the angular cut. On the heavy configuration this gives $\varepsilon = 6.3\,\text{mrad}$, fifteen times wider than the clean-configuration value, and section 6 shows what that width admits.

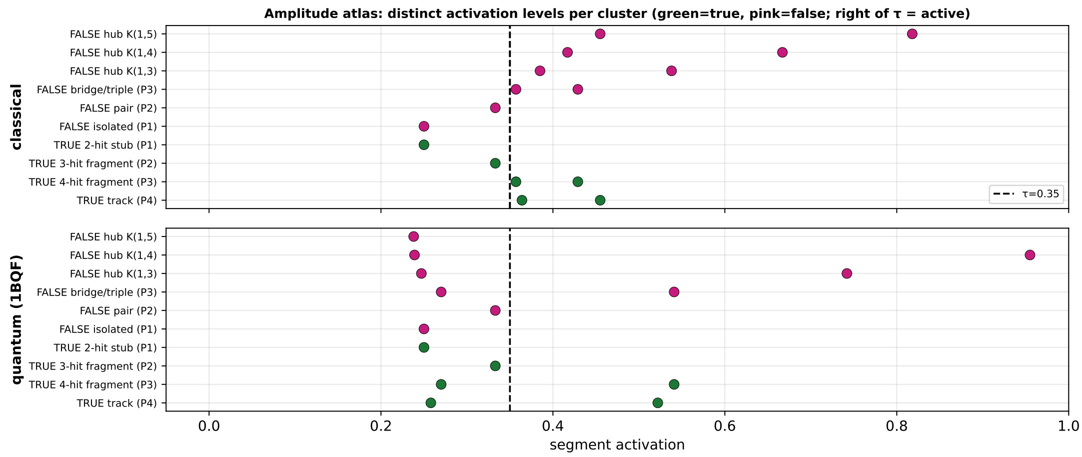


Figure 1: The amplitude atlas: exact activation level of every canonical cluster type (rows), classical (top) and 1BQF-filtered (bottom), with the fixed threshold $\tau = 0.35$ marked. Green: true; pink: false. Everything quantitative in this paper is a statement about where these levels sit relative to a threshold.

2.3 Exact activation levels: the atlas

Because A has constant diagonal, its action decomposes over the connected components of the compatibility graph, and small components can be solved by hand. This is worth doing once in full, because these little closed forms are the vocabulary of every result in the paper.

An *isolated* segment (no compatible neighbour) has the scalar equation $(\gamma + \delta)x = \delta$, so

$$x_{\text{iso}} = \frac{\delta}{\gamma + \delta} = \frac{1}{4} = 0.250. \tag{4}$$

This is the false *attractor*: at any multiplicity, the overwhelming majority of the $4T(T-1)$ false segments are isolated and sit at exactly 0.250. A true track, by contrast, is a path P_4 of four segments (five hits), whose matrix is $sI - P_4$ with P_4 the path adjacency. Its solution can be written down exactly:

$$x_{\text{outer}} = \frac{4}{11} \approx 0.364, \qquad x_{\text{inner}} = \frac{5}{11} \approx 0.455, \tag{5}$$

where the outer levels belong to the two end segments (one continuation neighbour each) and the inner levels to the two middle segments (two neighbours each). More generally, a chain of m segments has outer activation

$$x_{\text{outer}}(m) = \frac{1}{2} - \frac{r + r^m}{2\,(1 + r^{m+1})}, \qquad r = 2 - \sqrt{3}, \tag{6}$$

which climbs the ladder $\{0.250,\ \frac{1}{3},\ \frac{5}{14},\ \frac{4}{11}, \dots\} \nearrow 0.366$ for $m = 1, 2, 3, 4, \dots$ It is worth working two rungs by hand. A two-hit fragment (m=1) sits at exactly the isolated-false attractor 0.250, and a three-hit fragment (m=2) sits at 1/3, only 0.083 above it. Fragments are barely distinguishable from noise by amplitude alone; this innocent-looking fact returns as the fragment floor in section 9.

The classical decision rule is a threshold: segment i is *active* iff $x_i > \tau$ with

$$\tau(\gamma, \delta) = \frac{\delta}{\gamma + \delta} + 0.10, \tag{7}$$

i.e. $\tau = 0.35$ at $\gamma = 3$: a fixed margin above the false attractor, below the lowest clean-track level 4/11. The margin 0.10 is convention, inherited from ref. [3] and kept for comparability.

2.4 Metric definitions

Every number in this paper is one of the following, and I will never use these words in any other sense:

- **segment efficiency** $\varepsilon_{\text{seg}} = n_{\text{true active}}/n_{\text{true}}$: the fraction of true segments that survive selection;
- **segment false rate** $f_{\text{seg}} = n_{\text{false active}}/n_{\text{active}}$: the fraction of the *selected* sample that is fake (note the denominator: this is an impurity, not a false-positive rate over all fakes);
- **working points**: either the *fixed threshold* of eq. (7); or the *efficiency-first* point "wp99" (the threshold placed just below the 1% quantile of the true-segment amplitudes, so $\varepsilon_{\text{seg}} \geq 0.99$ by construction); or, in sections 8–9, the *matched-efficiency* convention: τ just below the k-th largest true amplitude with $k = \lceil \text{eff} \cdot n_{\text{true}} \rceil$, which lets solvers be compared at identical efficiency without quantile-grid artefacts.

Quantum solutions are unit-norm vectors; before thresholding they are rescaled to the classical amplitude scale on the classically-active support. Rescaling by the full norm would be multiplicity-dependent, since the false bulk owns most of the norm, and was the source of a metric artefact documented in the programme's earlier write-ups. All thresholds act on $|x_i|$.

2.5 The spectrum

Because $A = sI - C$, its spectrum is a rigid reflection of the compatibility graph's: $\lambda(A) = s - \mu(C)$. Three families matter:

1. **Isolated segments**: $\lambda = s$ exactly. This single line carries $\mathcal{O}(n_{\text{seg}})$ modes: at $T = 200$ under heavy noise, 155,371 of 157,609 segments (rep 0).
2. **Chains** P_m: $\lambda_k = s - 2\cos(k\pi/(m+1))$, $k = 1, \ldots, m$. A true track ($m = 4$) therefore lives on exactly four lines,
$$\lambda \in \{s - 2\cos(k\pi/5)\}_{k=1}^{4} = \{2.382,\ 3.382,\ 4.618,\ 5.618\} \quad (s = 4), \tag{8}$$
symmetric about s and split by the golden-ratio cosines[2].
3. **Stars** $K_{1,m}$ (one hit shared by m segments of the same role): $\lambda = s \pm \sqrt{m}$ plus $m-1$ copies of s.

The pair motif (m=1) at $\lambda = s \pm 1$ and the triple (m=2) at $\{s,\ s \pm \sqrt{2}\}$ do most of the false-population work under noise, and their lines interleave the true lines of eq. (8) without touching them. That gap, false lines close to but not on the true lines, is what this paper exploits. Motifs whose spectrum coincides with a true motif's are the exception, and they return in sections 5.5 and 9 as the limiting factor.

3 The one-bit quantum filter, read as a spectral filter

3.1 The circuit and its response function

The 1BQF of ref. [3] is a one-ancilla interference experiment. Prepare the uniform superposition $|u\rangle$ over the (zero-padded) segment register, put the ancilla in $(|0\rangle + |1\rangle)/\sqrt{2}$, apply e^{iAt} to the register controlled on the ancilla, close the interferometer with a second Hadamard, and keep the runs in which the ancilla reports the interference branch. The kept state is

$$|\psi\rangle \;\propto\; \tfrac{1}{2}(I + e^{iAt})|u\rangle \;=\; \sum_i e^{i\lambda_i t/2} \cos\!\Big(\frac{\lambda_i t}{2}\Big)\, \langle v_i|u\rangle\, |v_i\rangle, \tag{9}$$

[2] The values $2\cos(\pi/5) = \varphi = 1.618\ldots$ and $2\cos(2\pi/5) = 1/\varphi$: the five-hit track is secretly a pentagon problem.

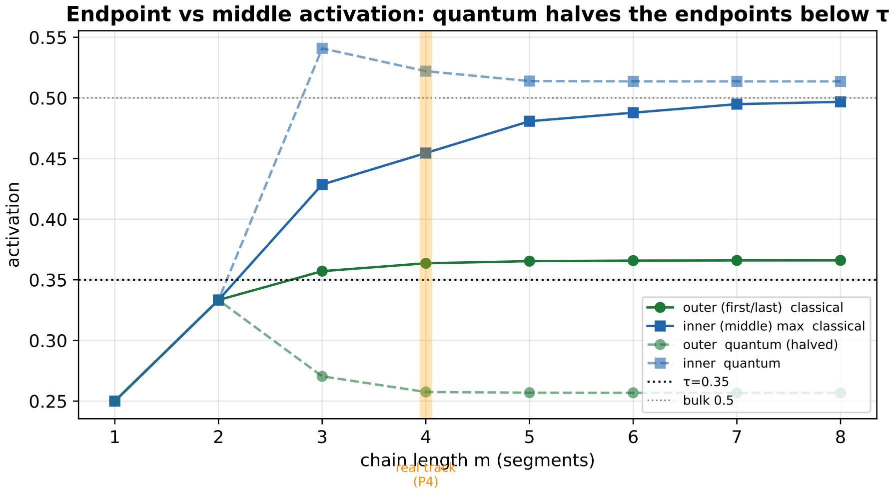


Figure 2: The chain ladder and the endpoint halving. Classical outer activation climbs eq. (6) toward 0.366 (above τ); the 1BQF response drops the P_4 outer level to 0.258, below the fixed threshold. This is the origin of the fixed-τ efficiency plateau, and the reason 1BQF headlines use the efficiency-first working point.

where $\{\lambda_i, |v_i\rangle\}$ are the eigenpairs of A. Reading out $|\langle j|\psi\rangle|$ segment by segment gives amplitudes x_j^Q, and the whole device is summarised by a single real response function applied to the spectrum:

$$f(\lambda) = \cos\Big(\frac{\lambda t}{2}\Big), \qquad t = \frac{\pi}{s}. \tag{10}$$

The choice $t = \pi/s$ places the zero of the cosine exactly at $\lambda = s$, the isolated-segment line. Every isolated false segment, all $\mathcal{O}(n_{\text{seg}})$ of them, is annihilated exactly: not suppressed, killed. This is the entire content of the 1BQF, and the form in which we will need it later is a simple accounting rule: each application of the filter buys exactly one zero of the response, and the 1BQF spends its one zero on the largest false population in the problem. The padding modes introduced to reach a power-of-two dimension are diagonal-only, sit at $\lambda = s$ too, and are removed by the same zero, a small but pleasing piece of engineering hygiene.

3.2 What the notch cannot do

Away from its zero the cosine is not flat, and this has two consequences that define the 1BQF's operating envelope.

The first is that true amplitudes are reshaped. A true track's four modes (eq. (8)) sit symmetrically about s, where $|f(\lambda)|$ is small; the filtered chain no longer reproduces the classical $\{4/11, 5/11\}$ profile. The effect has a closed form: the outer (endpoint) segments of a clean track drop from 0.364 to 0.258, below the fixed threshold $\tau = 0.35$, while the inner pair survives above it. At the fixed threshold the 1BQF therefore reports a segment efficiency plateau of ≈ 0.73–0.75: it finds the middles of tracks and loses the ends. This endpoint halving is a property of the cut, not of the physics; the endpoint amplitudes remain perfectly well ordered above the false ones. The honest convention, used for all 1BQF headlines in this paper, is the efficiency-first working point wp99: place the threshold just below the 1% quantile of true amplitudes and pay whatever false rate results. On clean fixed-acceptance events at $\gamma = 3$ that price is steep at high multiplicity: the wp99 false rate climbs to several tens of percent by $T = 400$ and beyond, and section 5 quantifies this against the comb.

The second consequence is that coupled false segments pass. A false segment that accidentally couples to the graph moves off the $\lambda = s$ line, out of the zero, and into the passband. The notch has no second zero to spend on it. As density or acceptance width grows, the coupled-false population grows, and the 1BQF's false rate grows with it. The 1BQF is, precisely, the statement that unconnected fakes are exactly removable; everything in this paper beyond section 4 is about the connected ones.

On success probability: the kept-branch probability scales as $P_{\text{anc}} \approx 0.25/T$ on this problem, since the uniform input spreads over $4T^2$ segments of which order T carry surviving amplitude, so shot budgets grow linearly in T and amplitude amplification buys the usual quadratic discount.[3] These costs carry over essentially unchanged to the polynomial filters below, which is why I defer resource accounting to section 5.4.

4 From one bit to any polynomial: the QSVT construction

The promotion from eq. (10) to an arbitrary polynomial response rests on three standard ingredients: block encoding, qubitization, and a linear combination of unitaries (LCU). I assemble them here in the concrete, real-arithmetic form used by the simulations. Readers who trust the QSVT literature [6–9] can skim to the resource summary in section 4.3; the details are included because the paper's cost claims (qubit counts, walk calls, success probabilities) are read directly off this construction.

4.1 Block encoding and the qubitization walk

Rescale the Hamiltonian affinely so its spectrum fits in $[-1, 1]$: $X = (A - c_0 I)/c_1$ with c_0 the spectrum centre and c_1 slightly more than its half-width[4]. A block encoding is a unitary that hides X in its top-left corner:

$$U = \begin{pmatrix} X & \sqrt{I - X^2} \\ \sqrt{I - X^2} & -X \end{pmatrix}, \tag{11}$$

acting on the segment register plus one *dilation* qubit. The picture I find most useful is a rotation in each eigenplane: for every eigenvalue x of X there is a two-dimensional invariant subspace in which U rotates by the angle $\arccos x$. The *qubitization walk* $W = (Z \otimes I)\, U$ turns these into clean rotations with the defining property

$$\langle 0|\, W^k\, |0\rangle_{\text{dil}} = T_k(X), \tag{12}$$

where T_k is the k-th Chebyshev polynomial and the bra-ket projects the dilation qubit onto $|0\rangle$. Equation (12) is exact, with no Trotterization and no error terms, and every operator in it is real, so the entire simulation can run in real arithmetic on orthogonal matrices. Walking k steps and looking at the undeflected component applies the k-th Chebyshev polynomial of the Hamiltonian; that is the whole theorem.

4.2 Summing the series: LCU

A degree-d response $p(X) = \sum_{k=0}^{d} c_k T_k(X)$ is then a weighted sum of walk powers. Introduce an index register of $m = \lceil \log_2(d{+}1) \rceil$ qubits, prepare $|\text{prep}\rangle \propto \sum_k \sqrt{|c_k|}\, |k\rangle$, apply W^k controlled on the index (in practice: controlled powers W^{2^j}), undo the preparation, and post-select the index and dilation registers on zero. The surviving register state is $\propto p(X)|u\rangle$ with success probability

$$P_{\text{anc}} = \frac{\big\| p(X)\, |u\rangle \big\|^2}{\big(\sum_k |c_k| \big)^2}, \tag{13}$$

[3] The 0.25 constant is the analytic estimate for the uniform input; the comb's analogous constant, ≈ 0.44, is directly measured (section 5.4).

[4] The implementation pads the user-supplied spectral bounds by a few percent; an early campaign bug taught us that the polynomial's fit domain must exceed the padded bounds or the constructor rightly refuses.

the usual LCU price: the one-norm of the coefficient vector. The 1BQF is recovered, exactly, as the $d = 1$ row of this table (cos is T_1 of the appropriately rescaled operator), which is the formal content of the claim that this paper continues rather than replaces its predecessors.

Simulation-side, eq. (12) gives a matrix-free evaluator: $p(A)\mathbf{b}$ by the Chebyshev three-term recursion on sparse matrix-vector products, $\mathcal{O}(d \cdot \mathrm{nnz}(A))$ flops, which runs in about 1.6 s at $T = 1000$ ($n = 4 \times 10^6$) and agrees with the full qiskit circuit and with a gate-streaming evaluation to 10^{-9} where all three are affordable. All large-T results below use this exact evaluator; nothing in this paper relies on an approximate simulation of the quantum readout.

4.3 Resource summary

For a degree-d filter on an event of T tracks:

- **Width**: $n_{\text{sys}} = \lceil \log_2 4T^2 \rceil$ system qubits $+1$ dilation $+m = \lceil \log_2(d+1) \rceil$ index qubits. With the production degree $d = 40$ this is 21 qubits at $T = 50$ rising to 29 at $T = 1000$. Width is cheap, and grows only logarithmically.
- **Depth**: d controlled walk calls per coherent repetition, times $\mathcal{O}(1/\sqrt{P_{\text{anc}}})$ amplitude-amplification rounds.
- **Success**: $P_{\text{anc}} \propto 1/T$ (measured constant ≈ 0.44 for the comb of section 5), so total walk calls scale as $d\sqrt{T}$.

The quantum content of this paper is this resource profile; the response functions themselves are classically simulable by the same recursion, and I return to what that does and does not concede in section 10.5.

5 Fitting the polynomial to the clean events: the line comb

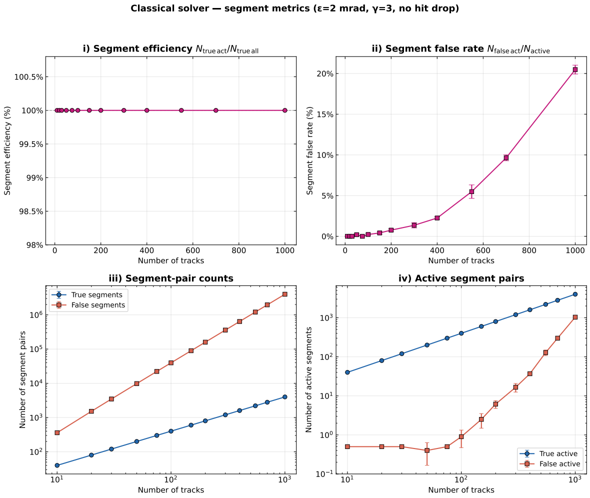
5.1 The design insight: the true spectrum is discrete

Here is the observation that turns QSVT from a mathematical possibility into a design tool. On this detector geometry every true track has exactly five hits, hence exactly four segments, hence, by eq. (8), exactly the same four eigenvalues. The true spectrum of the entire event, however many thousand tracks it contains, is four discrete lines. We are not trying to pass a region of the spectrum; we are trying to pass four known radio stations, and everything between them is static.

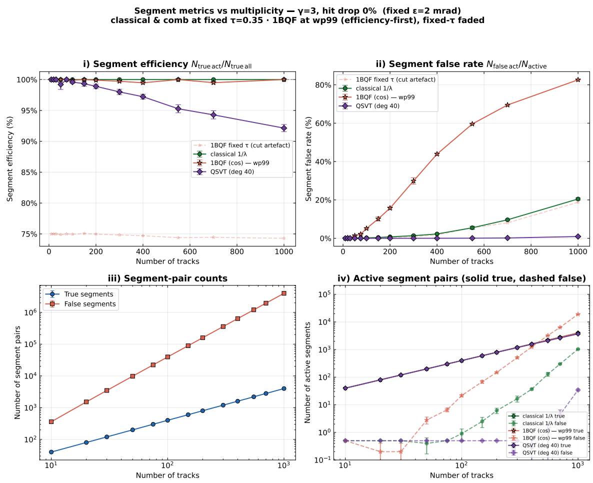
My first design did not understand this. A band-limited approximate inverse ($\approx 1/\lambda$ across a contiguous band containing the true lines, zero outside) works beautifully at low multiplicity and then fails exactly where it matters: as T grows, accidental tangles of false segments produce eigenvalues throughout the band, and a band design passes them all. At fixed threshold its false rate reached tens of percent by $T = 400$, worse than classical.[5] The production design is therefore a *line comb*: narrow Gaussian passes of half-width h_w, weighted $\propto 1/\lambda$, centred at the four lines $s - 2\cos(k\pi/5)$ only, fitted as a bounded Chebyshev series of degree d. The defaults, fixed once by a two-dimensional (d, h_w) scan and never tuned per event, are $d = 40$, $h_w = 0.18$.

Why does a width of 0.18 work? The nearest systematically-populated false lines, the P_3 bridge modes at $\{s - \sqrt{2},\ s,\ s + \sqrt{2}\}$, sit $\gtrsim 0.2$ away from the true lines, so the comb threads the gap; and the resolution law of a degree-d Chebyshev series, features no narrower than $\approx 2W/d$ on a domain of half-width W, makes $d = 40$ sufficient for $h_w = 0.18$ on the clean-event domain.

[5] A useful failure: it taught us that the exploitable structure is the discreteness of the geometry-pinned true spectrum, not its location.



(a) Classical solver.



(b) Line-comb QSVT.

Figure 3: The clean-toy 2×2 ($\gamma = 3$, fixed $\varepsilon = 2\,\mathrm{mrad}$, no hit drop): segment efficiency, false rate, and pair counts against multiplicity, classical (left) versus the production line comb (right). The corresponding 1BQF panels appear in the study records; its fixed-τ/wp99 duality is described in section 3.

Degree, width, and line spacing are one budget, and this is the budget the noisy events of section 9 will re-open.

5.2 Clean-event results

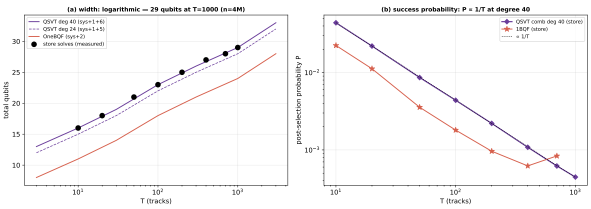
On the fixed-acceptance ($\varepsilon = 2\,\mathrm{mrad}$), $\gamma = 3$ study, the canonical benchmark of the programme, with events and solutions shared segment-for-segment across all three solvers through a common data pipeline, the picture at the fixed threshold $\tau = 0.35$ is:

- **Classical**: efficiency 100% throughout (the minimum true amplitude 4/11 clears τ at every T); false rate grows from ≈ 0 at low T to 2.25% at $T = 400$ and 20.5% at $T = 1000$ as the $\mathcal{O}(T^2)$ false pairing feeds the active set.
- **1BQF**: the fixed-τ efficiency plateau of section 3 (74.7% at $T = 400$, the endpoint cut); false rate at or just below classical (1.73% at $T = 400$, 18.8% at $T = 1000$).
- **Comb**: efficiency 97–100% with false rate $\leq 0.02\%$ through $T = 400$ (zero to a few false actives per event where the classical solver admits tens), rising to 0.17% at $T = 700$ and 0.9% at $T = 1000$ against the classical 20.5%, at efficiency 92.1%.

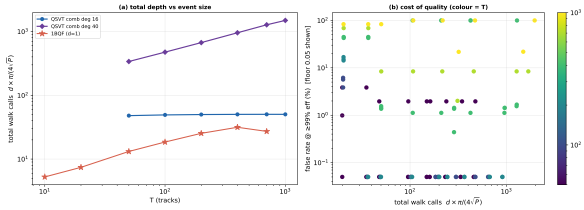
At the efficiency-first wp99 point the comb wins clearly through $T = 400$ (1.3% vs classical 3.1% on matched fresh solves), but at extreme density the composition effect dominates everyone: the false-to-true ratio $n_F/n_T \approx T - 1$ converts even excellent per-segment separation into a growing impurity, and by $T = 1000$ the wp99 false rates cross ($\approx 21\%$ comb vs 17.5% classical, crossover near $T = 700$).[6] The two regimes, fixed-τ where the comb wins by an order of magnitude and more, and wp99 at extreme density where composition flattens everyone, bracket the operational truth.

The comb's amplitude picture explains the numbers. Under the comb response the false population collapses to $\lesssim 10^{-2}$ of the true median (the passes reject everything off the lines), while the true band stays tight around its classical profile: the $1/\lambda$ weighting inside the passes preserves the chain shape rather than reshaping it as the single cosine does. Where the 1BQF splits the true

[6] The crossover is robust; its size is aggregation-dependent. A whole-store aggregate reads the classical wp99 false rate at $T = 1000$ as low as 10%. Both sources agree the comb no longer wins at wp99 beyond $T \approx 700$, which is the operative statement.



(a) Register width and success probability.



(b) Total walk calls (depth).

Figure 4: Measured resource scaling of the comb on the shared events: width grows logarithmically (21 qubits at $T = 50$ to 29 at $T = 1000$); success probability falls as $\approx 0.44/T$; amplified walk calls grow as $d\sqrt{T}$, with the half-resolved $d = 16$ comb holding a T-independent ≈ 50-call budget to $T = 700$.

population and the classical solver drags a false tail across the threshold, the comb simply leaves a gap.

5.3 Hit inefficiency: the first appearance of fragments

Dropping 1% of hits breaks $\approx 5\%$ of tracks into fragments: chains of one, two, or three segments whose spectra (section 2.5) are *not* on the four P_4 lines. The comb, by design, does not pass them. Per-class efficiency under the production comb is 100% on intact tracks and 0% on every fragment class, which is why its overall efficiency reads 94.0% rather than 97–98% at $T = 400$ under drop. Extending the comb with passes at the fragment lines does not help: it re-admits their false spectral twins wholesale (a P_2 fragment and a false pair are the same matrix), so fragments are a spectral floor, the first appearance of the twin obstruction that section 9 meets at full strength. Recovery, if wanted, must come from information a spectral filter cannot see, namely hit-level constraint information, and that is classical post-processing, outside the scope of this paper.[7]

5.4 Resource scaling and the depth ladder

Measured on the shared events: total width 21 qubits at $T = 50$ to 29 at $T = 1000$; success probability $P_{\text{anc}} \approx 0.44/T$; walk calls per amplified repetition $d \cdot \pi/(4\sqrt{P_{\text{anc}}}) \propto d\sqrt{T}$. Two findings sharpen the budget. A half-resolved comb at $d = 16$ matches the $d = 40$ metrics up to $T = 700$ at an approximately T-independent $P_{\text{anc}} \approx 6\%$, about fifty total walk calls, flat in T, because partially-merged passes still cover the four lines while rejecting the bulk. And the degree axis is not monotonic: intermediate degrees ($d \approx 20$–32) can cliff at the efficiency-first working point through Chebyshev fit pathologies, so every production degree must be validated, not interpolated.

5.5 Provable limits on the clean toy

Three structural results bound what any response function can do here. The first is the *floor theorem*: if a false cluster's matrix is graph-isomorphic to a true cluster's (a false "bridge" of exactly four segments, say), then for every function f, $f(A)\mathbf{b}$ assigns both clusters identical amplitudes, because permutation similarity commutes with matrix functions and $\mathbf{b}$ is uniform. Such topological twins are invisible to the entire filter family, classical scoring included. They are rare on clean events (none found in the $T \leq 200$ census; 0.6% of true segments by $T = 400$ with the acceptance widened to 4 mrad) but they are the true ceiling. The second is the *shared-line theorem*: the odd modes of a contaminated five-segment chain coincide exactly with the $s \pm \sqrt{3}$ lines of the three-pronged hub $K_{1,3}$, so no comb can null both populations independently; pass

[7] An earlier version of this study paired a hit-occupancy recovery gate with the comb alone; it was retired as mission creep (no other solver received the same help, and the tool is orthogonal to the spectral question). Classical post-selection is developed and reported separately in the programme's occupancy study.

menus trade one against the other with a fixed exchange rate. The third is the *mirror tax*: a chain-reflection symmetry forces the false extension of a recovered cluster to the recovered-true amplitude level, so recovery passes always buy their efficiency at a false-rate premium. Together these three results closed the clean-toy design question. The production comb is optimal up to provable exceptions, and further progress must come from changing the operator or leaving the spectral domain, which is exactly where the rest of this paper goes.

6 The heavy-noise operating point

From here on the operating point is the heavy configuration: $\sigma_{\text{scatt}} = 10^{-4}$, $\sigma_{\text{res}} = 20\,\mu\text{m}$, hit drop 1%, at $T = 200$, with the acceptance set by the kink formula to $\varepsilon = 6.3$ mrad. Three reps of every event are used throughout; where a single event is dissected, it is rep 0.

The acceptance width is the story. At $\varepsilon = 2$ mrad a false segment rarely finds an angular partner; at 6.3 mrad the compatibility graph fills with small false structures. Two facts about that graph, both established by direct decomposition of the solved events, control everything downstream:

1. **There is no giant tangle.** The compatibility graph at heavy $T = 200$ splits into ≈ 870 nontrivial connected components of at most ≈ 17 segments, floating in a sea of 155,371 isolated segments (rep 0). Noise, at this operating point, buys many small motifs rather than one big mess, and small components mean the entire spectrum is computable exactly, component by component, with dense eigensolvers. Every spectral statement in sections 7–9 is exact, not approximate.
2. **The false actives are motif-borne.** At the classical wp99 point the event has 2238 active segments of which 1454 are false; those false actives live in 707 components that contain, between them, only 142 true segments. Ninety percent of the false-active population sits in essentially pure-false motifs (pairs, triples, short bridges), not entangled with true tracks at all.

The solver scoreboard at this point makes uncomfortable reading. At wp99 the classical false rate is 0.65; the 1BQF's is 0.65 as well, since the notch's one zero still annihilates the isolated bulk while every false *active* is by definition coupled, off the line, and untouched. The production comb, designed for a 2 mrad world where false structures barely exist, fares worst of all: the wide acceptance populates false motif lines throughout its pass windows. The measured false-active pile-up (section 9) will show exactly where. The important statement here is the diagnosis: at heavy noise all three solvers fail for the same reason, a false population that has moved into spectral regions the designs were never told about. The classical literature offers two Hamiltonian terms as candidate cures, occupancy and bifurcation. The next two sections take them in turn, and both turn out to behave in ways their classical reputations do not predict.

7 The occupancy term

7.1 The term

Denby's original network [4] constrains each hit to be used by at most one selected segment per role. The quadratic form of that constraint, per-hit occupancy, adds $E_{\text{occ}} = \alpha \sum_h (o_h - 1)^2$ with $o_h = \sum_{s \ni h} x_s$, which at the matrix level reads

$$A_{\text{occ}} = A + 4\alpha I + 2\alpha B_{\text{all}}, \qquad \mathbf{b} \to \mathbf{b} + 4\alpha \mathbf{1}, \tag{14}$$

where B_{all} couples every pair of segments sharing a hit in the same role: two segments departing the same hit, or two arriving at it. Consecutive segments of one track share their interior hit in opposite roles and are not coupled. The isolated attractor moves to $(\delta + 4\alpha)/(\gamma + \delta + 4\alpha)$ and the threshold tracks it.

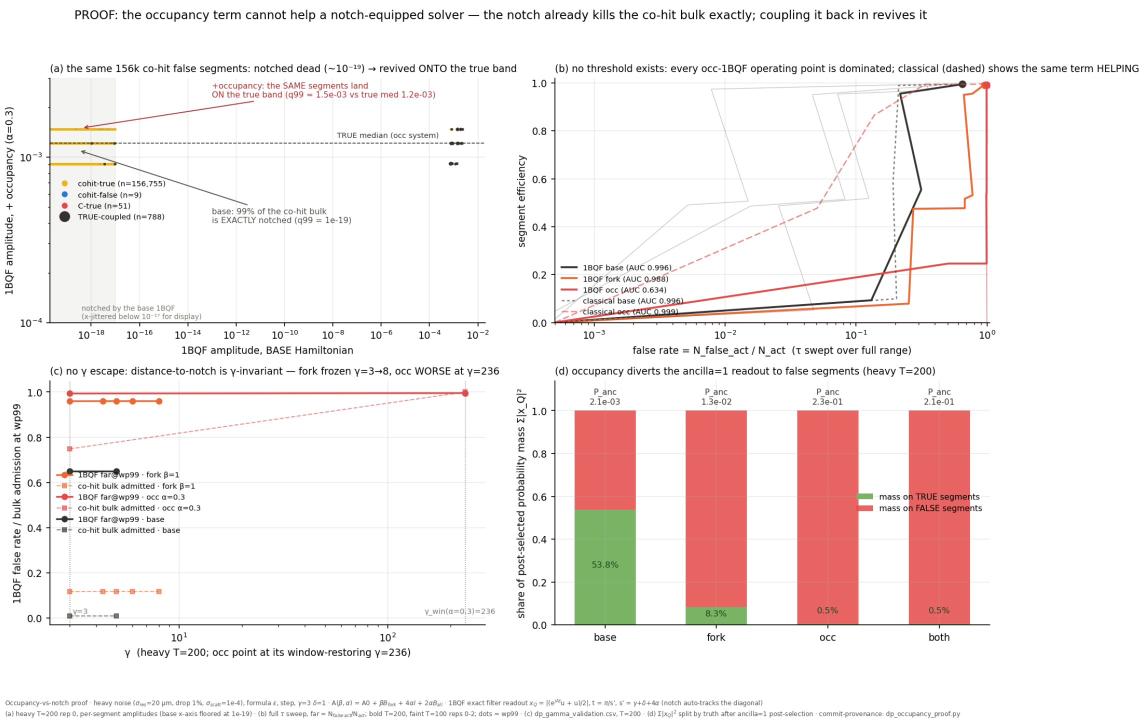


Figure 5: The occupancy proof (heavy $T = 200$): (a) the same co-hit false segments, exactly notched under the base Hamiltonian, re-emerge on the true amplitude band under A_{occ}; (b) no threshold escape: every occupancy-1BQF operating point is dominated, while the same term helps the classical solver; (c) no γ escape; (d) the post-selected probability mass diverts to false segments.

7.2 Classically it works; spectrally it is a trap

On the heavy $T = 200$ benchmark event the in-matrix occupancy term at $\alpha = 0.3$ does exactly what Denby promised: the classical wp99 false rate falls from 0.650 to 0.210 at efficiency 0.991. If this paper were classical, the section would end here with a recommendation.

For the 1BQF it is a disaster, and the mechanism is worth spelling out because it generalises. The notch's power is that isolated false segments sit exactly on its zero. The occupancy coupling $2\alpha B_{\text{all}}$ connects precisely those segments to their co-hit neighbours, which is its job, and thereby moves them off the zero. On the benchmark event the co-hit false bulk, notched to amplitudes $\sim 10^{-19}$ under the base Hamiltonian (99% of it below 10^{-17}), re-emerges under A_{occ} at amplitudes on the true band: its 99th percentile rises to 1.5×10^{-3} against a true median of 1.2×10^{-3}. The rank separation collapses (1BQF AUC 0.996 $\to$ 0.634) and no threshold rescues it; the full sweep shows every occupancy-1BQF operating point dominated by the base-1BQF curve. Nor does the self-coupling γ offer an escape: the notch tracks the diagonal, distance-to-notch is γ-invariant, and at the window-restoring $\gamma = 236$ the occupancy system is strictly worse. Post-selection tells the same story: the occupancy term diverts the ancilla-conditioned probability mass almost entirely onto false segments.

The general lesson, stated once here and used again for the fork, is this: a spectral filter earns its living from populations pinned to known eigenvalues, and any Hamiltonian term that couples a pinned population unpins it. Constraint terms that help an amplitude-reading solver can therefore destroy a spectrum-reading one. The occupancy constraint is real physics, since hits really are used once, but if it is to be used at all it must act where it cannot smear the spectrum: after the solve, as classical post-processing.[8] For this paper the occupancy verdict is the negative

[8]The programme develops that post-processing tool, a hit-uniqueness selection applied evenly to every solver's output, in a separate occupancy study; it is deliberately kept out of this paper, whose subject is the spectral side.

result, and it is worth stating plainly: as a Hamiltonian term, occupancy is poison to a notch, and the classical intuition for the term does not transfer.

8 The fork term and the moving spectrum

8.1 The term and its validity envelope

The second Denby–Peterson penalty targets *bifurcations*: two segments claiming the same hit in the same role at small mutual angle, the fork topology that produces the closely-competing candidates a wide acceptance admits. The ε-windowed form adds

$$A' = A + \beta\, B_{\text{fork}}(\varepsilon_B), \tag{15}$$

where B_{fork} couples co-hit same-role pairs within angular window ε_B (default: the acceptance itself). Unlike the occupancy term, the fork is surgical where it matters. On clean events it touches $\approx 2\%$ of segments, and provably no pure-true chain carries a fork edge, so the true comb lines are β-invariant. Earlier classical-side work in this programme established that the fork migrates contaminated-true spectral weight back toward the comb bands, and an emulated composition projected an operating point near 99% efficiency at sub-percent false rate, the target this section set out to validate with real solvers.

Adding a positive semi-definite coupling widens the spectrum, and the solver imposes two validity conditions with closed forms. Positive definiteness requires $\gamma > \gamma_{\text{pd}} = \gamma_{\text{ref}} - \lambda_{\min}$; the one-period condition of any cosine-family filter (the top of the spectrum must not alias through the response period) requires $\gamma > \gamma_{\text{win}} = \lambda_{\max} - \gamma_{\text{ref}} - 2\delta$, both extremes measured at a reference γ_{ref} and shifted rigidly, since γ enters only the diagonal. At heavy $T = 200$ the measured requirement is $\gamma^* \approx 3.4$ at $\beta = 0.5$ and ≈ 4.4 at $\beta = 1$. The notch tracks the diagonal automatically ($t = \pi/s'$, $s' = \gamma + \delta$), and the comb is rebuilt on the measured shifted spectrum. Every fork result below carries these retunes; a window-violating control ($\beta = 0.5$ at $\gamma = 3$) is kept in the study to show they matter.

8.2 The measured no-go with the unfitted comb

Run exactly so, with the *production* comb, the fork strictly hurts. The exact-readout 1BQF's wp99 false rate at heavy $T = 200$ climbs $0.65 \to 0.73 \to 0.96$ for $\beta = 0 \to 0.5 \to 1$; the comb's wp99 reading is pinned near 0.995 for every β (a convention degeneracy, discussed below); and the emulated target is nowhere in sight. Taken at face value this is a no-go for the operator knob on the quantum side.

Two observations kept the verdict from being final. The first is that the wp99 convention degenerates for the comb at heavy noise: contaminated true segments carry near-zero comb amplitude, the 1%-quantile threshold collapses to its floor, and the resulting false-rate reading of 0.995 describes the convention, not the filter. This is why section 9 switches to matched-efficiency comparisons. The second observation is the decisive one: the comb was never refitted to the fork system's false population. The retunes above move the comb's centre and domain; they do not ask where the false weight now lives. The no-go, in other words, was a verdict on an unfitted filter, and the correct response was not to abandon the operator knob but to complete the design loop it opens. Change the Hamiltonian, and the activations and their spectral support move; when they move, the polynomial must be refitted to the moved spectrum. Carrying out that refit is the subject of section 9.

8.3 A Trotter aside that will matter on hardware

The same study ran the real product-formula OneBQF circuit against the exact evolution on identical fork systems, and found a clean split. With uniform couplings ($\beta = 0$, and $\beta = 1$ where

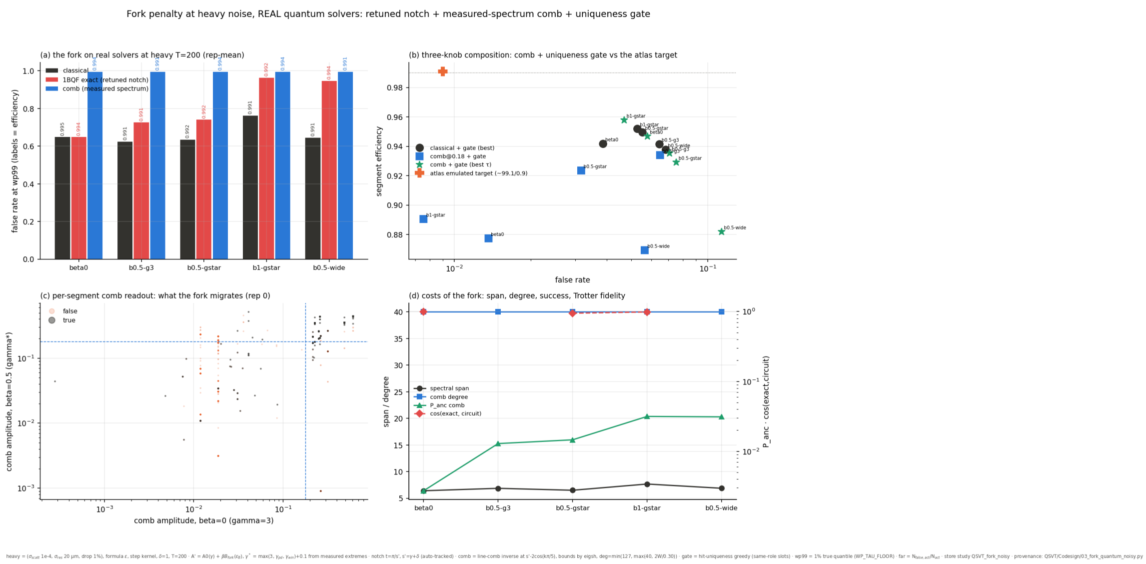


Figure 6: The fork with the *unfitted* comb (heavy $T = 200$): wp99 false rates per solver and β (a); compositions against the emulated target (b); per-segment comb readout migration (c); spectral span, degree, success and circuit fidelity costs (d).

fork edges match compatibility edges in weight) circuit and exact evolution agree to $\cos \approx 0.985$; at heterogeneous $\beta = 0.5$ the per-pair product formula scatters the readout ($\cos = 0.944$, circuit false rate 0.96 against the exact 0.73). An independent arbiter run on the erf angular-cost Hamiltonian, same events, exact e^{iAt} against the stored circuits, shows the identical pattern at larger amplitude: mixed-weight kernels are Trotter-dominated (false rate $0.83 \to 0.22$ at moderate noise once the product formula is removed), uniform-weight kernels are not. The rule of thumb for hardware is that heterogeneous coupling weights break the naive product formula, and circuit realisations of weighted Hamiltonians need the structured synthesis route (edge-coloured ordering) developed in the programme's direct-structural-synthesis study. All fork and fit results in this paper use exact evolution precisely so that filter physics and circuit error are never conflated.

9 Fitting the polynomial to noisy events: roots at the measured pile-up

9.1 Measuring the false pile-up

The fit begins with the measurement the no-go was missing. Because the heavy-noise compatibility graph has no giant component (section 6), the per-component eigendecomposition gives, for every eigenvalue in the event, its exact content: how much of the mode's weight lies on true segments, how much on false, and how much of each segment's amplitude it builds. Under any response p,

$$x_s(p) = \sum_i p(\lambda_i)\,(\mathbf{v}_i^\top \mathbf{b})\,v_i(s), \tag{16}$$

linear in p, so the question of where the false amplitude comes from is well posed and exactly answerable. The answer, on the heavy $T = 200$ events, is concentrated to a degree I did not expect. The false-active formation weight piles up on the *pair line* $\lambda = s - 1$: on the three events measured, that single bin carries 1122–1378 weight units against 4–10 units of true weight, with the triple lines $s \pm \sqrt{2}$ next; the overall bin-level overlap between false-active and true formation is 0.014–0.023 in cosine. The false population is not smeared. It is parked on the eigenvalue lines of identifiable two- and three-segment motifs, almost none of which are shared with true formation.

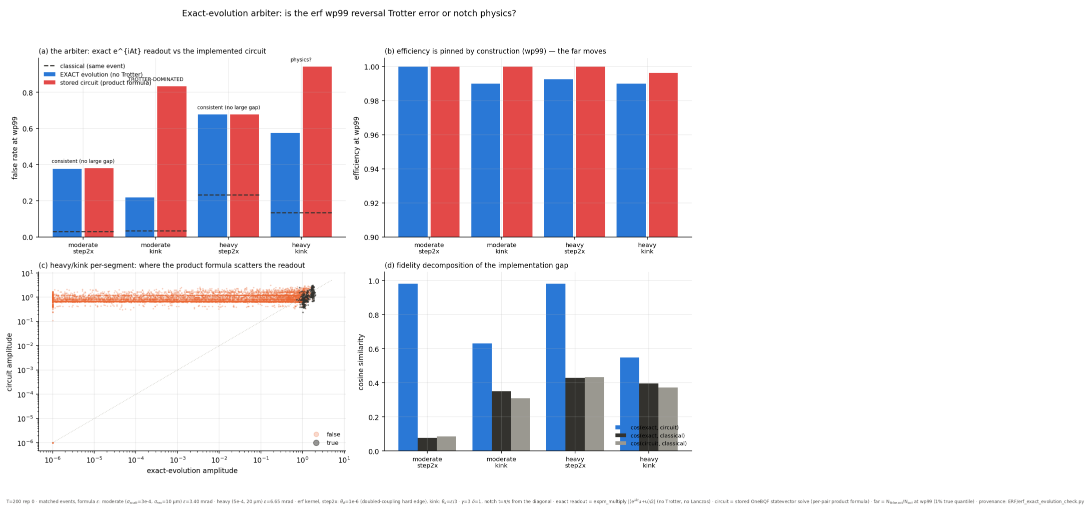


Figure 7: The exact-evolution arbiter on the erf angular-cost Hamiltonian: uniform-coupling kernels (step2x) show circuit = exact (the deficit is physics); finite-width kernels (kink) are Trotter-dominated. Removing the product formula collapses the false rate from 0.83 to 0.22 (moderate) and 0.94 to 0.58 (heavy).

9.2 Fitting the response

With eq. (16) linear in the response, the fit is a regularised least-squares problem over binned response values ($\Delta\lambda = 0.02$):

$$\min_{p} \sum_{s\,\in\,\text{false}} x_s(p)^2 \;+\; \mu \sum_{s\,\in\,\text{true}} \left(x_s(p) - x_s^{\text{cls}}\right)^2, \qquad |p| \le 1, \tag{17}$$

with the isolated-false line pressured to zero and μ swept to trace the efficiency–purity trade. No root position is chosen by hand: zeros appear where false weight concentrates because that is what the objective rewards. The fitted response reproduces the comb's passes at the (contamination-shifted) true lines and drives deep roots through $s - 1$ and $s \pm \sqrt{2}$, which is precisely the polynomial the no-go was missing.

9.3 Results

Comparisons are at matched efficiency (the k-th-true-amplitude convention of section 2.4), rep-averaged over three events per configuration. The main table, heavy noise, $T = 200$:

config	eff	classical	1BQF	comb (fixed)	**fitted**
$\beta = 0$	0.98	0.124	0.199	0.733	**0.096**
$\beta = 0$	0.97	0.124	0.199	0.466	**0.061**
$\beta = 0.5, \gamma^*$	0.98	0.463	0.145	0.960	**0.085**
$\beta = 0.5, \gamma^*$	0.97	0.292	0.124	0.960	**0.046**
any	0.99	0.63–0.76	0.65–0.96	$\approx$1	0.63–0.77

Table 1: Segment false rate at matched efficiency, heavy noise, $T = 200$ (rep-means). "Fitted" is the optimal binned response of eq. (17).

Three conclusions follow.

First, the expected ordering is restored. Below efficiency ≈ 0.985 the fitted quantum filter sits under the classical baseline everywhere: 0.096 against 0.124 at $\varepsilon_{\text{seg}} = 0.98$, 0.061 against 0.124 at $\varepsilon_{\text{seg}} = 0.97$. The claim this section set out to test, that the false rate should still be lower for the quantum filters once they are fitted, holds.

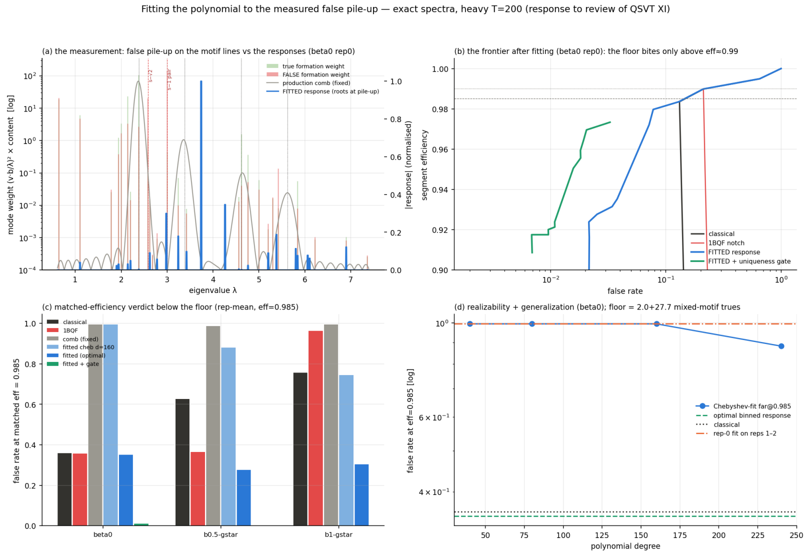


Figure 8: Fitting the response to the measured pile-up (heavy $T = 200$): (a) exact per-mode false (red) and true (green) formation weight against λ, with the fixed comb and the fitted response overlaid; the fitted roots land on the pair and triple lines. (b) The resulting frontier, with the fragment floor biting only above $\varepsilon_{\text{seg}} \approx 0.99$. (c) Matched-efficiency verdict across configurations. (d) Realizability and generalization of the fitted response.

Second, the fork is redeemed. On the fork system the fitted response reaches 0.046 at $\varepsilon_{\text{seg}} = 0.97$, beating the fitted no-fork system's 0.061, while the classical solver on the same fork system reads 0.292. The operator knob does exactly what the spectral-migration analysis promised, provided the response is refitted; paired with a fixed response it reads as a no-go. Changing the Hamiltonian without refitting the polynomial is not a half-measure, it is worse than doing neither. The standard-notch 1BQF also benefits from the fork below the floor (0.145 against 0.199 at $\varepsilon_{\text{seg}} = 0.98$): the fork pushes coupled-false modes away from the passband even when the only zero sits at s'.

Third, the wall at $\varepsilon_{\text{seg}} \geq 0.99$ is real, universal, and now has a name. At the strict efficiency-first point everything, classical included, floods back to $f_{\text{seg}} \approx 0.65$. The per-component census locates the cause exactly: $\approx 2\%$ of *true* segments (directly counted: 15 and 22 of 783/788 on two of the three events[9]) are spectral twins of the false motifs. They are the hit-drop fragments of section 5.3: a broken track's two-segment piece *is* a pair, sitting on the pair line, in a pure-true component, plus a handful of contaminated mixed chains. Any response with a root at $s - 1$ kills them alongside the false pairs; their population exceeds the 1% efficiency budget; and once the working point must dip below their (suppressed) amplitudes, the entire suppressed false band comes back above threshold. This is the *fragment floor*: the noisy-event generalisation of the clean floor theorem, no longer a curiosity but the binding constraint, and no function of λ, fitted or otherwise, can cross it. Recovering those segments requires hit-level information, which is classical track-level work outside the spectral domain and outside this paper.

9.4 Why the 1BQF specifically cannot follow

A degree-one filter owns one zero, and the measurement says the heavy false spectrum has two mandatory root locations: the isolated line at s ($\mathcal{O}(10^5)$ segments) and the pair line at $s - 1$

[9] On the third event the census heuristic reads 52 through a scale-sensitive leak-tier definition (its fit selected a tenfold-larger regulariser), and it degenerates on the fork configurations, so I quote only the directly-counted events and flag the third as unresolved. The robust, heuristic-free statement is the measured wall itself: every solver holds $\varepsilon_{\text{seg}} = 0.98$ and none holds $\varepsilon_{\text{seg}} = 0.99$, on all three events.

(the active pile-up). The study's control makes the conflict vivid. Retuning the notch onto the pair line ($t = \pi/(s-1)$) does root the pile-up, and releases the isolated bulk, which re-enters at amplitude $|\cos(st/2)|\,\delta = 0.5$, on top of the true band, for a false rate of 0.995. The 1BQF's zero is spoken for. Covering both lines is degree two; covering the triple lines as well is degree four and up; and that, stated as a root budget, is in my opinion the cleanest justification for the whole QSVT programme that the toy has produced.

10 Limitations

I collect here, deliberately and in one place, everything that stands between the results above and a deployable algorithm. Each limitation comes with its measured size and, where we have one, its escape route.

10.1 Realizability of the fitted response

The fitted optimum of eq. (17) has features of width ≈ 0.02 on a spectral span of ≈ 7. A single global Chebyshev series resolving that needs degree $d \approx 2W/h \approx 700$; our realizability ladder confirms the arithmetic (at $d = 240$ the realised filter still reads 0.88 false rate at $\varepsilon_{\text{seg}} = 0.985$, far from the optimum's 0.35 on the same convention). Global realization at the fitted resolution is therefore expensive, hundreds of walk calls per repetition. The escape is structural: the same no-giant-component fact that made the spectrum measurable makes the problem block-diagonal. Solving per connected component (a classical $\mathcal{O}(n)$ preprocessing step) replaces one $4T^2$-dimensional filter problem by hundreds of ≤ 17-dimensional ones, on which a degree-≤ 17 interpolating polynomial reproduces any response exactly, and as a bonus the register shrinks to $\lceil \log_2 17 \rceil + 2$ qubits and the success probability per cluster becomes $\mathcal{O}(1)$. The cost is honesty about what remains quantum: per-cluster solving helps the classical solver identically, and the quantum claim reduces to the resource profile of section 4.3 on whatever cluster sizes survive preprocessing.

10.2 Generalization of the fit

The response fitted on one event transfers poorly to its statistical siblings (0.63–0.99 false rate on cross-application, against 0.04–0.10 in-sample). The decomposition is instructive. The root positions are universal ($s-1$ and $s \pm \sqrt{2}$ are motif properties, not event properties), but the fitted pass positions chase each event's particular contaminated-true modes, which shift with the noise realisation. A production design must therefore be a universal response: roots pinned to the motif lines, passes wide enough to cover the contaminated-true spread measured across an event ensemble, fitted jointly rather than per event. The per-event fit of section 9 then plays the role its construction suggests, the measured ceiling that a universal design approaches. Building that universal response, and quantifying the ceiling gap, is the programme's clear next step.

10.3 Trotterization on hardware

Everything above uses exact evolution. On hardware, the 1BQF's product-formula circuit is exact only in the uniform-coupling case; the moment coupling weights differ (erf-weighted kernels, fractional fork strengths) the naive per-pair formula scatters the readout by amounts that dominate the physics (section 8). The structured-synthesis route (edge-coloured pair ordering, developed in the programme's DSS study, which also reaches 1BQF width for the comb via generalized QSP over e^{-iAt}) restores exactness at matched cost and is, on present evidence, mandatory for any weighted Hamiltonian on hardware.

10.4 Success probability and readout

The global filters run at $P_{\text{anc}} \approx 0.44/T$ (comb) and $\approx 0.25/T$ (1BQF): thousandths at $T = 1000$ before amplification. Amplitude amplification restores $\sqrt{T}$ scaling of total walk calls, and active-set readout (sampling until the $\mathcal{O}(T)$ surviving segments are each seen) gives the measured $d \cdot T^{1.5} \log T$ walk-call budget against the classical $d \cdot T^2$ flop budget: a square-root-class separation for that readout task, and no more than that.

10.5 Dequantisation honesty

Every response function in this paper can be applied classically by the Chebyshev recursion in $\mathcal{O}(d \cdot \text{nnz}(A))$ flops, seconds at $T = 1000$. The discrimination physics (combs, roots, floors) is solver-independent mathematics of the segment Hamiltonian, and this paper's contribution to it stands whether or not a quantum computer is ever involved. The quantum claim is confined to the resource profile: width logarithmic in problem size, walk calls $d\sqrt{T}$ against classical dT^2 for the readout task above, and a hardware-native route to the same amplitudes. Readers allergic to quantum advantage claims may read the whole paper as classical spectral-filter engineering with a quantum implementation appendix; nothing in the physics sections would change.

10.6 The toy's geometry, and what a real detector changes

Finally, the largest caveat of all: the four-line comb exists because this toy pins every true track to five hits. Real VELO tracks have variable length, and a companion study on real Run-3 event geometry found the true spectrum smeared into a band, the P_4 comb keeping only 40% of true segments, and a short-fragment floor ($m \leq 2$ tracks indistinguishable from noise at the segment level) capping all segment-level solvers, while the fork transfer and the band-plus-notch redesign behaved as the toy predicts. The honest statement of scope is that this paper's method is response engineering on whatever spectral structure the geometry provides, demonstrated exhaustively on a geometry that provides a lot of it; the measurement loop of section 9 (measure the pile-up, fit the response, locate the floor) is geometry-agnostic even where the specific comb is not.

11 Conclusion

This paper set out to finish a thought that the one-bit quantum filter started. The 1BQF's insight was that track finding does not need a linear-system solution, only a spectral separation, and that one cosine with one zero buys the largest separation in the problem at almost no quantum cost. Promoting that cosine to the full QSVT polynomial family converts algorithm design into response design, and response design turns out to be a measurement-driven loop:

1. Read the spectrum the problem gives you. On clean events it hands you four exact lines, and a fixed forty-degree comb converts them into a false rate twenty times below classical at $T = 1000$.

2. When the problem changes, measure where the false weight went, and put roots there. At heavy noise the pile-up sits on the pair and triple motif lines; the fitted response beats classical at every attainable efficiency, and the fork term, useless under a fixed response, becomes a net win under a refitted one. Operator and response are one design, not two.

3. Know what the spectrum cannot see. Spectral twins, false bridges on clean events and true fragments on noisy ones, bound every function of λ, classical scoring included; at heavy noise they define a fragment floor at $\varepsilon_{\text{seg}} \approx 0.98$. Constraint physics can close part of that gap, but only from outside the Hamiltonian: inside it, the same physics destroys the notch it was meant to help, which is the occupancy result of section 7.

The open problems are, I think, unusually crisp for this field: a universal (event-ensemble) response to replace the per-event fit; a per-cluster realization to replace the degree-700 global polynomial; edge-coloured synthesis for weighted couplings on hardware; and the transplant of the whole loop onto real detector geometry, where the first measurements say the method survives even though the comb does not. Every cloud has a silver lining: the same noise that broke the clean-event comb handed us a false population parked on nameable eigenvalue lines, and a filter family with exactly enough roots to meet it.

Acknowledgements

To be completed (predecessor-paper co-authors, supervision, computing resources at Nikhef).

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